

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Let $A_1, A_2, \dots, A_k \in \mathbb{C}^{m \times n}$ and let $\alpha \in \mathbb{C}$. Prove by induction on k that for $k \geq 2$,

$$\alpha(A_1 + A_2 + \dots + A_k) = \alpha A_1 + \alpha A_2 + \dots + \alpha A_k.$$

2. Let $A, B \in \mathbb{C}^{n \times n}$. Prove that $AB = BA$ if and only if $(A + B)(A - B) = (A - B)(A + B)$.

For full credit, you must cite all Laws of Matrix Transposes, Laws of Matrix Multiplication, and Laws of Matrix Addition and Scalar Multiplication that you use.

3. (a) Let $A, B \in \mathbb{R}^{m \times n}$. Suppose that $v^T A w = v^T B w$ for all $v \in \mathbb{R}^{m \times 1}$, $w \in \mathbb{R}^{n \times 1}$. Prove that $A = B$.

- (b) Give an example of two matrices $A, B \in \mathbb{R}^{2 \times 2}$ such that $v^T A v = v^T B v$ for all $v \in \mathbb{R}^{2 \times 1}$, but $A \neq B$.

4. Recall that a matrix B is called **symmetric** if $B^T = B$. Similarly, a matrix C is called **skew-symmetric** if $C^T = -C$. Let $A \in \mathbb{R}^{n \times n}$ be a square matrix with real entries.

- (a) Show that the matrix $B = \frac{1}{2}(A + A^T)$ is symmetric.
- (b) Show that the matrix $C = \frac{1}{2}(A - A^T)$ is skew-symmetric.
- (c) Show that any square matrix A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.
- (d) Show that $x^T A x = x^T B x$ for any vector x such that the product is defined, where B is as defined in part (a).

For full credit, you must cite all Laws of Matrix Transposes, Laws of Matrix Multiplication, and Laws of Matrix Addition and Scalar Multiplication that you use.



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- From *Linear Algebra with Applications* by Nicholson

- Section 2.1: 2, 3, 4, 11, 12, 19, 22, 23b
- Section 2.2: 3, 15, 21
- Section 2.3: 1, 3, 31, 34a, 35
- Appendix B: B.1, B.2, B.4, B.5a, B.5b
- Appendix C: C.1, C.2, C.21

1. Give an example of two nonzero matrices $A \in \mathbb{R}^{2 \times 3}$ and $B \in \mathbb{R}^{3 \times 2}$ such that $AB = 0$.
2. Let $\mathbf{v}$ and $\mathbf{w}$ be $n \times 1$ vectors. Prove or disprove (with a counterexample) each of the following statements.

(a) $\mathbf{v}^T \mathbf{w} = \mathbf{vw}^T$

(b) $\mathbf{vw}^T = \mathbf{wv}^T$

(c) $\mathbf{v}^T \mathbf{w} = \mathbf{w}^T \mathbf{v}$

